



Tetracycline degradation by persulfate activated with magnetic Cu/CuFe₂O₄ composite: Efficiency, stability, mechanism and degradation pathway

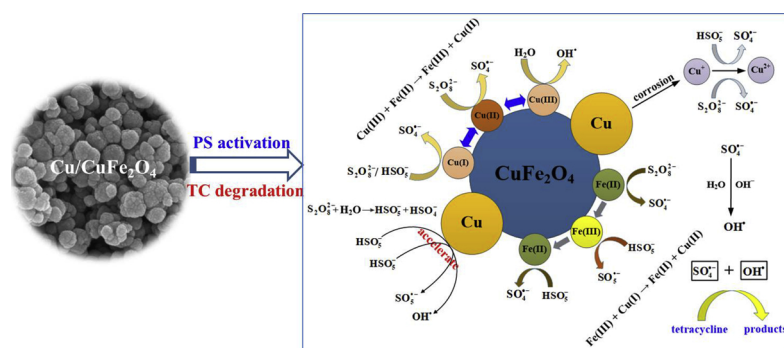


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GRAPHICAL ABSTRACT



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ABSTRACT

Magnetically separable Cu/CuFe₂O₄ composite obtained by a solvothermal method was used to activate persulfate (PS) for the removal of tetracycline (TC). Under different pH conditions, Cu/CuFe₂O₄ catalyst exhibited a higher catalytic activity for PS activation to degrade TC than that of CuFe₂O₄. The effects of some key parameters including initial pH value, PS concentration, catalyst dosage, reaction temperature and coexisting ions on TC degradation were investigated in Cu/CuFe₂O₄/PS system. The reuse of Cu/CuFe₂O₄ catalyst at pH 3.50, 7.00 and 11.00 indicated that the catalyst showed a low stability due to the corrosion of metallic copper (Cu⁰), but bicarbonate ions could enhance the stability and recyclability of this catalyst through the suppression of copper leaching. Both sulfate and hydroxyl radicals were the main reactive species in Cu/CuFe₂O₄/PS system. Cu⁰ can not only work as electron donor to active PS to produce the reactive radicals but also act as an electron bridge to facilitate the fast electron transfer between PS and catalyst. The structural cuprous and ferrous ions on the surface of CuFe₂O₄ participated in the PS activation process through the redox reactions, as confirmed by XPS analysis. The possible degradation pathways of TC were proposed based on the identified intermediates.

1. Introduction

Tetracycline (TC), a category of antibiotic widely used to treat bacterial infectious disease, has been found to interrupt the balance of

harmonious coexistence between nature and human beings since it has been detected in various compartments (surface water, drinking water and soil) [1–4]. Moreover, TC cannot be thoroughly degraded by the conventional wastewater treatment processes due to its antibacterial

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and refractory properties [5–7]. It is urgently essential to develop an efficient remediation technology to remove TC.

Fenton-like advanced oxidation processes, which utilized oxidants (i.e., H_2O_2 , peroxymonosulfate (PMS) and persulfate (PS)) to generate the reactive species, have attracted considerable attention for wastewater purification [7–9]. Compared with H_2O_2 or PMS, PS is convenient to transport, low-cost, much more persistent and stable [10]. Besides, PS can be activated to generate $\text{SO}_4^{\cdot-}$ radicals with a higher oxidation potential and a longer half-life (2.5–3.1 V, 30–40 μs) than $\text{OH}^{\cdot}$ radicals (2.2–2.7 V, < 1 μs) [11]. As reported, the catalysts used for PS activation included Fe/C, CuO, Fe_3O_4 , $\text{Fe}_3\text{O}_4/\text{C}$, $\text{CuO}/\text{Fe}_3\text{O}_4$ and CuFe_2O_4 [8,12–14]. In particular, CuFe_2O_4 has attracted considerable research attention owing to its excellent properties, such as the recoverability and mechanical stability [15–20]. Li et al. evaluated the removal of *p*-nitrophenol (PNP) in $\text{CuFe}_2\text{O}_4/\text{PS}$ system, and they found that neutral condition showed the best removal efficiency and basic solution was not suitable for PNP degradation [14]. Zhang et al. used carbon nanotubes/ CuFe_2O_4 composite to activate PS for diethyl phthalate (DEP) degradation. The removal of DEP was strongly dependent on pH, and the best degradation efficiency of DEP was achieved at acidic condition [21]. Therefore, an effective strategy should be pursued to enhance the activity of CuFe_2O_4 to activate PS for the removal of organic pollutants under different pH conditions.

Recently, zero valent metals (ZVMs) have been found to exhibit the remarkable catalytic activity for organic pollutants degradation, such as Fe^0 [22], Al^0 [23], Ni/Fe [24] and Cu^0 [25,26]. Among them, Cu^0 was often used in the PS activation process. For example, Zhou et al. employed the Cu^0/PS system to remove diverse contaminations from water, and they found that PS could promote corrosion of Cu^0 to release Cu^+ ions, which further activated PS to generate reactive species [27]. Ji et al. reported that Cu^0 in Fe/Cu bimetallic particles could improve the activity of Fe^0 for PNP degradation due to its rapid electron transfer in accelerating the corrosion of Fe^0 [9]. The similar results were also observed by Xiong et al. [28] and Lai et al. [29]. Obviously, Cu^0 can not only activate PS to generate the reactive species but also promote the catalytic processes involving electron transfer. Thus, it is feasible to improve the catalytic activity of CuFe_2O_4 for PS activation through combining CuFe_2O_4 with Cu^0 .

Herein, $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite was prepared by a simple solvothermal route. $\text{Cu}/\text{CuFe}_2\text{O}_4$ coupled with PS can effectively degrade TC over a wide pH range. Besides, the effect of some key operating parameters on TC degradation in $\text{Cu}/\text{CuFe}_2\text{O}_4/\text{PS}$ system was investigated. Specially, the recyclability and stability of $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite can be enhanced in the presence of bicarbonate ions. The mechanism of $\text{Cu}/\text{CuFe}_2\text{O}_4$ activated PS for TC degradation was elucidated in detail. Finally, the possible degradation pathways of TC were proposed according to the identification of intermediates.

2. Experimental section

2.1. Chemicals

All the chemical reagents were listed in Text S1.

2.2. Preparation of $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite

The $\text{Cu}/\text{CuFe}_2\text{O}_4$ catalyst was synthesized according to the previous work [30]. In brief, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and NaAc were mixed in 30 mL of ethylene glycol, and then heated at 200 °C for 12 h. The precipitate was separated with a magnet, washed and dried at 80 °C. Without adding FeCl_3 , Cu^0 was synthesized through the above process. CuFe_2O_4 was obtained by removing Cu^0 in $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite using FeCl_3 aqueous solution [31]. The characterization of as-obtained catalysts was provided in Text S2.

2.3. TC degradation experiments

TC degradation experiments were carried out in a 250 mL glass beaker and the temperature was controlled at 25 °C. Firstly, 30 mg of catalyst was added into TC solution (100 mL, 50 mg/L) with desired initial pH value (pH_0) and then the desired amount of PS was added into the above TC solution to start the reaction under mechanical stirring. At determined time intervals, 2.0 mL of the suspension was collected and followed by immediately adding 1.0 mL of methanol to inhibit TC from further degradation [10].

2.4. Analysis

The concentration of TC was analyzed using a UV spectrophotometer (UV-2000, Mapada) at a wavelength of 359 nm. The total copper ions released into solution were measured by an atomic absorption spectroscopy (TAS-990, Pgeneral). The pH values of the samples were detected using a pH meter (PHS-25, China). The mineralization of TC was measured by a TOC Analyzer (TOC-L CPH, Shimadzu). The sulfate and hydroxyl radicals were identified by an electron paramagnetic resonance spectrometer (EPR, Bruker E500). The degradation intermediates of TC were detected by HPLC-MS/MS system (Waters Xevo TQ-S) with a Waters BEH C18 column (2.1 × 50 mm, 10 cm) (Text S3).

3. Results and discussion

3.1. Characterization of the catalysts

Fig. S1 displays the XRD patterns of the as-obtained catalysts. Obviously, the $\text{Cu}/\text{CuFe}_2\text{O}_4$ sample exhibits the coexistence of Cu^0 and CuFe_2O_4 . Fig. 1a shows the SEM image of $\text{Cu}/\text{CuFe}_2\text{O}_4$ sample. It can be seen that $\text{Cu}/\text{CuFe}_2\text{O}_4$ catalyst consists of sphere-like particles with the diameter of 50–200 nm. The TEM image of $\text{Cu}/\text{CuFe}_2\text{O}_4$ catalyst indicates that the sphere-like particles are assembled by small nanoparticles (Fig. 1b). The lattice fringes of CuFe_2O_4 and Cu^0 are 0.48 and 0.25 nm (Fig. 1c), which are assigned to the crystal planes of CuFe_2O_4 ($\bar{1}\bar{1}\bar{1}$) and Cu^0 (110), respectively.

XPS measurement was carried out to detect the chemical composition of $\text{Cu}/\text{CuFe}_2\text{O}_4$ catalyst. Fig. 2a shows the existences of C, Cu, Fe, and O in $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite, and C is probably ascribed to the adventitious hydrocarbon contaminants [32]. Fig. 2b displays the six contributions fitted in Cu 2p XPS spectrum originating from Cu $2\text{p}_{3/2}$, Cu $2\text{p}_{1/2}$ and satellite peaks of Cu^{2+} [33]. The satellite peaks at 942.60 and 939.53 eV are indexed to the dominant copper oxide species [34,35]. The peaks at 953.7 and 933.7 eV correspond to Cu^{2+} , and the peaks located at 951.8 and 932.0 eV can be indexed to Cu^+ [14,31]. Besides, according to the area ratio of $\text{Cu}^{2+}/\text{Cu}^+$, the mass percentage of Cu^0 in $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite was calculated to be about 32.6%. Fig. 2c shows the six peaks divided from the Fe 2p XPS spectrum, in which the peaks at 725.64 and 712.39 eV with a satellite signal at 718.20 eV are assigned to Fe^{3+} , and the peaks at 723.16 and 709.96 eV confirmed the presence of Fe^{2+} with mass proportion calculated to be about 63.3% [32]. XPS spectrum of O 1s is shown in Fig. 2d, in which one peak at 529.7 eV is ascribed to lattice oxygen O and the other peak at 530.8 eV can be attributed to the surface hydroxyl groups [32,36].

The *M-H* hysteresis loop of $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite shows that the saturation moment is 54 $\text{emu} \cdot \text{g}^{-1}$ (Fig. S2a). The $\text{Cu}/\text{CuFe}_2\text{O}_4$ catalyst can be totally gathered within 2 s due to its strong magnetic ability, and it can also be easily dispersed into the solution with a slight shaking after the magnet was taken away (Fig. S2b).

3.2. TC degradation

3.2.1. TC degradation by different catalysts

As shown in Fig. 3, PS alone cannot oxidize TC at pH_0 3.50 and 7.00,

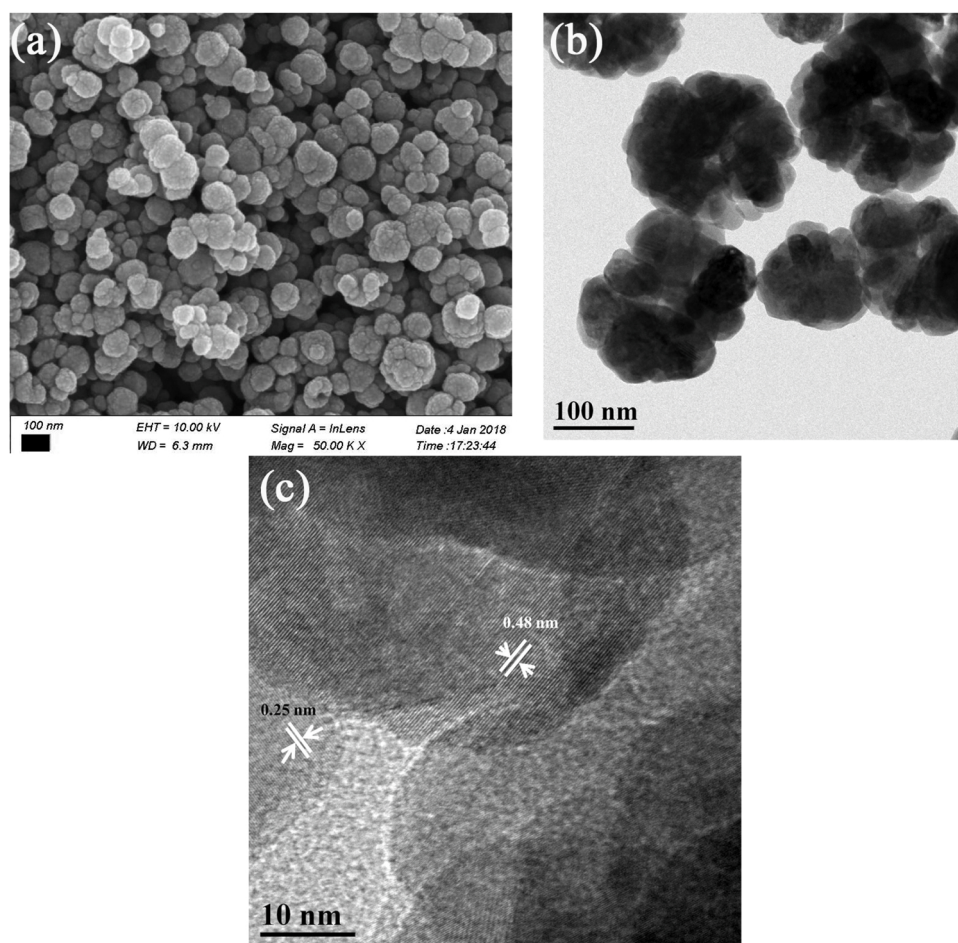


Fig. 1. SEM image (a), TEM (b) and HRTEM image (c) of CuFe_2O_4 composite.

and 60% of TC was degraded at pH_0 11.00 because the basic condition could increase the reactivity of PS to form reactive radicals [37]. Obviously, the as-obtained catalysts can effectively activate PS to degrade TC at different pH_0 (Fig. 3). Previous work reported that the higher surface area and larger pore volume would be beneficial for the contact between the catalyst and PS and lead to a higher reactivity [38]. The specific surface area and pore volume of $\text{Cu/CuFe}_2\text{O}_4$ composite are smaller than pure CuFe_2O_4 (Table 1), however, $\text{Cu/CuFe}_2\text{O}_4$ catalyst exhibits a higher catalytic activity than that of CuFe_2O_4 at different pH_0 (Fig. 3). This result indicated that Cu° played a significant role in the PS activation process. Pure Cu° was found even more reactive than $\text{Cu/CuFe}_2\text{O}_4$ composite for TC degradation in the first 60 min (Fig. 3a and b) due to the corrosion of Cu° to generate a large amount of Cu^+ ions, which could active PS to produce reactive species [25–27]. Actually, it can be clearly observed that Cu° catalyst can be totally dissolved after 120 min of reaction in $\text{Cu}^\circ/\text{PS}$ system (Fig. 3d). At different pH_0 , the concentrations of leached copper ions in different systems are displayed in Fig. 3d. Almost no copper ions were leached in $\text{CuFe}_2\text{O}_4/\text{PS}$ system after 120 min. The concentration of leached copper ions is about 100 mg/L in $\text{Cu/CuFe}_2\text{O}_4/\text{PS}$ system, and about 300 mg/L in $\text{Cu}^\circ/\text{PS}$ system (Fig. 3d). Although the concentration of leached copper ions in $\text{Cu}^\circ/\text{PS}$ system is much higher than that in $\text{Cu/CuFe}_2\text{O}_4/\text{system}$, TC removal efficiency in $\text{Cu/CuFe}_2\text{O}_4/\text{system}$ is slightly higher than that in $\text{Cu}^\circ/\text{PS}$ system after 120 min of reaction (Fig. 3). This result confirmed that TC degradation in $\text{Cu/CuFe}_2\text{O}_4/\text{PS}$ system proceeded through a homogeneous catalytic process and a heterogeneous catalytic reaction. Furthermore, the adsorption of TC by $\text{Cu/CuFe}_2\text{O}_4$ under different pH conditions was also investigated. After adsorption for 120 min, the TC removal efficiency was 18%, 14% and 6% at pH_0 3.50, 7.00 and 11.00,

respectively (Fig. S3). The pH_{pzc} of $\text{Cu/CuFe}_2\text{O}_4$ catalyst was about 7.45 (Text S4 and Fig. S4). $\text{Cu/CuFe}_2\text{O}_4$ is positively charged when solution pH is below 7.45, and is negatively charged as pH is above 7.45. In the pH range of 3.3–7.7, TC presents as zwitterionic species [6]. Thus, TC adsorption by $\text{Cu/CuFe}_2\text{O}_4$ exhibits a slight decrease from pH 3.50 to 7.00. TC presents as anionic species when pH is higher than 7.7 [6]. The electrostatic interaction between $\text{Cu/CuFe}_2\text{O}_4$ and TC is repulsive at pH_0 11.00, hence, only 6% of TC was adsorbed by $\text{Cu/CuFe}_2\text{O}_4$ catalyst (Fig. S4). The FT-IR spectra of TC, $\text{Cu/CuFe}_2\text{O}_4$ catalyst and $\text{Cu/CuFe}_2\text{O}_4$ after adsorption are shown in Fig. S5. It can be observed that the FT-IR spectrum of $\text{Cu/CuFe}_2\text{O}_4$ catalyst did not change after TC adsorption under different pH conditions. Obviously, one can deduce that TC removal in $\text{Cu/CuFe}_2\text{O}_4/\text{PS}$ system was mainly attributed to the catalytic degradation process (Fig. 3 and Fig. S3).

It should be noted that the recycle of $\text{Cu/CuFe}_2\text{O}_4$ is not favorable because of its relatively high copper ions leaching after the corrosion of Cu° in the oxidation process (Fig. 3d). In addition, although copper is not regarded as a potential carcinogen [17], much more copper ions discharged into natural water would have some influence on ecological environment. A way of reducing the leaching of copper ions in $\text{Cu/CuFe}_2\text{O}_4/\text{PS}$ system will be proposed in the following discussion.

3.2.2. Effect of solution pH

The performances of PS-based advanced oxidation processes were strongly dependent on the solution pH [14,34,38,39]. Obviously, negligible changes in the removal efficiency of TC were observed in a wide pH range from 3.50 to 11.00 (Fig. 4a). To further investigate the influence of pH, the variation of solution pH with time elapse was monitored. Obviously, the solution pH was immensely decreased

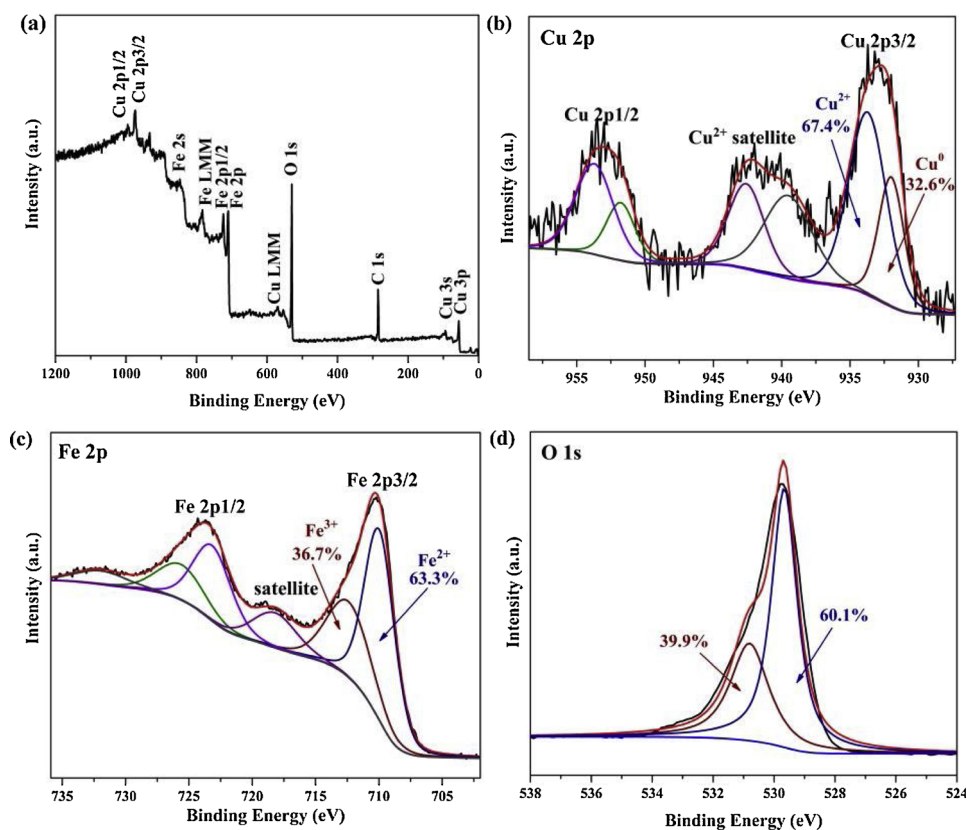


Fig. 2. XPS spectra of Cu/CuFe₂O₄ composite: (a) survey spectra; (b) Cu 2p; (c) Fe 2p; (d) O 1s.

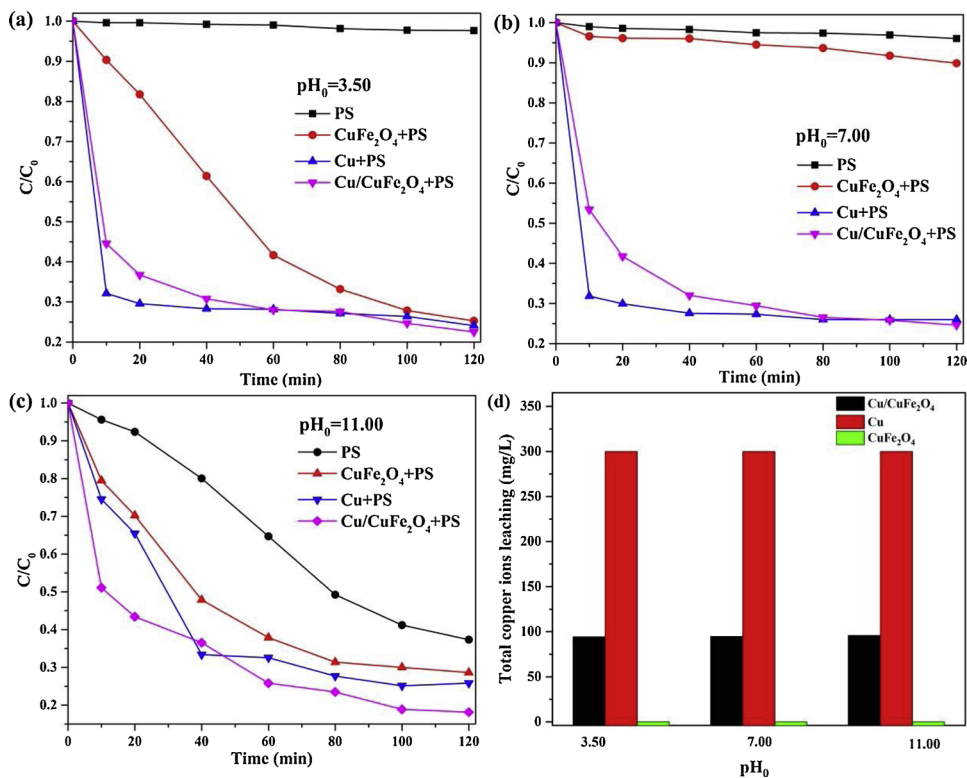


Fig. 3. TC degradation and the leaching of copper ions in the different systems. Conditions: [PS] = 1.5 g L⁻¹, [TC] = 50 mg L⁻¹, [catalyst] = 0.3 g L⁻¹.

Table 1
The BET surface area and pore volume of the as-obtained catalysts.

sample	BET surface area ($\text{m}^2 \text{g}^{-1}$)	pore volume (cm^3/g)
CuFe_2O_4	24.82	0.1918
$\text{Cu}/\text{CuFe}_2\text{O}_4$	13.98	0.021

throughout TC degradation processes in $\text{Cu}/\text{CuFe}_2\text{O}_4/\text{PS}$ system, especially in neutral or basic conditions (Fig. S6). This phenomenon was due to the decomposition of PS by $\text{Cu}/\text{CuFe}_2\text{O}_4$, the formation of HSO^- through the reaction between PS and H_2O (Eq. (1)) [34,38], and the generation of low molecular weight acids (Table S1, $m/z = 118$, $m/z = 73$). Besides, the lower pH is beneficial for the corrosion of Cu^+ , and responsible for the enhanced activity [25,27]. The degree of TC mineralization in $\text{Cu}/\text{CuFe}_2\text{O}_4/\text{PS}$ system is shown in Fig. 4b. After 120 min of reaction, the TOC removal efficiency was 59.1%, 23.4%, 24.0%, 29.1% and 68.3% at pH_0 3.50, 5.81, 7.00, 9.44, and 11.00, respectively. The maximum TOC removal was obtained at pH_0 11.00

because the basic condition can promote the reactivity of PS [37].



3.2.3. Effect of PS concentration

As displayed in Fig. 4c, the removal efficiency of TC increased with increasing PS concentrations. To be specific, 66.7%, 72.9% and 75.4% of TC was eliminated with increasing PS concentration from 0.30 g/L to 1.50 g/L. The accelerated generation of $\text{SO}^{\cdot-}$ radicals was responsible for the faster TC degradation at higher PS concentration [40]. However, the degradation efficiency of TC slightly decreased to 72.8% when the PS concentration increased to 2.25 g/L. This result may be due to the quenching effect of $\text{SO}^{\cdot-}$ radicals by PS itself [21,41].

3.2.4. Effect of catalyst dosage

After 120 min, the degradation efficiency of TC continuously increased from 71.8% to 75.4% with increasing $\text{Cu}/\text{CuFe}_2\text{O}_4$ dosage from 0.1 g/L to 0.3 g/L (Fig. 4d). This could be due to the fact that the total surface area and active sites on the surface of $\text{Cu}/\text{CuFe}_2\text{O}_4$ composite increased as $\text{Cu}/\text{CuFe}_2\text{O}_4$ dosage increasing, which was beneficial for

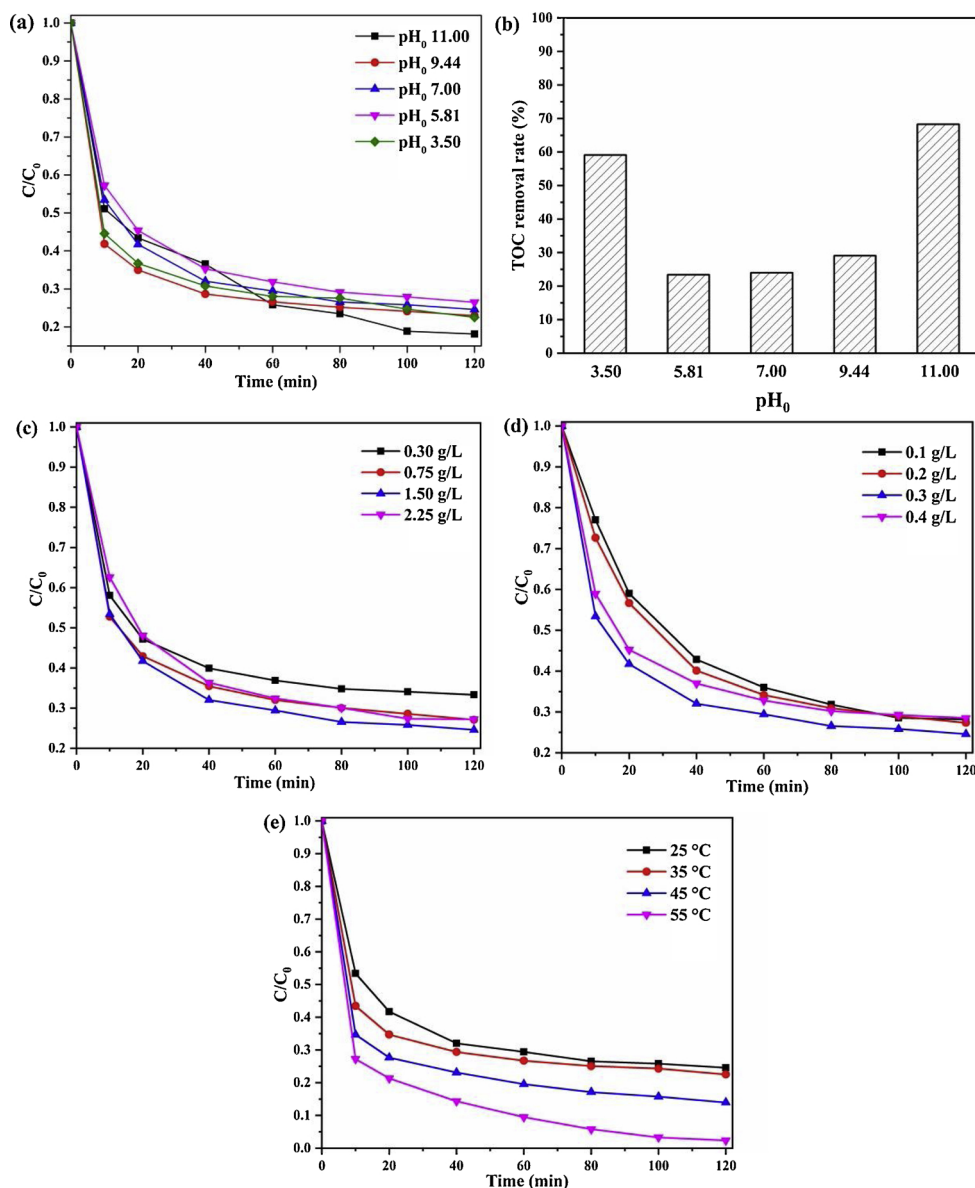


Fig. 4. (a) The influence of pH_0 on TC degradation; (b) The TOC removal efficiency in $\text{Cu}/\text{CuFe}_2\text{O}_4/\text{PS}$ system at different pH_0 ; The effect of PS dosage (c), catalyst dosage (d) and temperature (e) on TC removal.

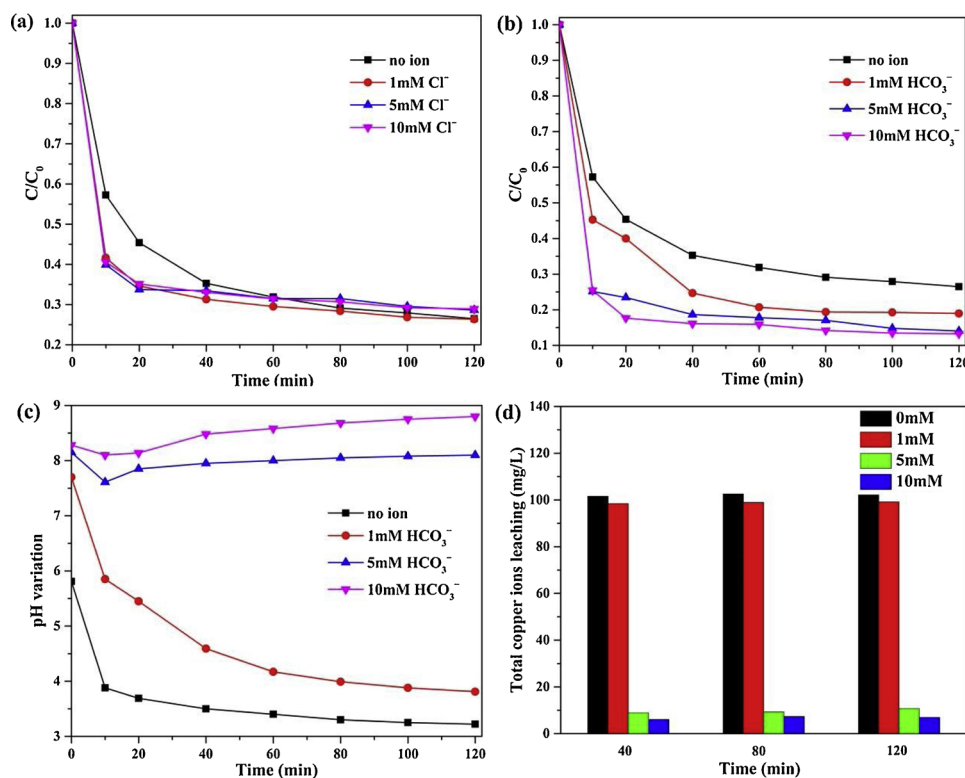


Fig. 5. The effect of inorganic anions on TC degradation: (a) Cl^- ; (b) HCO_3^- ; (c) pH variation and (d) total copper ions leaching in the presence of HCO_3^- . Conditions: $[PS]_0 = 1.5 \text{ g L}^{-1}$, $[TC] = 50 \text{ mg L}^{-1}$, $[Cu/CuFe_2O_4] = 0.3 \text{ g L}^{-1}$.

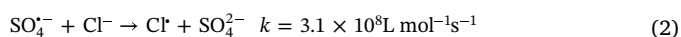
activating PS to produce $SO^{\cdot-}$ 4radicals [14]. However, a mild decrease was found when the catalyst dosage increased to 0.4 g/L (Fig. 4d) because the excess catalyst would quench the reactive radicals [38,39].

3.2.5. Effect of temperature

Fig. 4e shows the influence of the reaction temperature on TC degradation, and the results indicated that heat activation was beneficial for TC removal. After 120 min, the removal efficiency of TC gradually increased from 75.4% to 77.4%, 86.0% and 97.6% when the temperature increased from 25°C to 35 , 45 and 55°C , respectively. In general, elevated temperature can promote the electron transfer from $Cu/CuFe_2O_4$ catalyst to PS to generate reactive radicals [10,42,43], thereby improving the removal efficiency of TC.

3.2.6. Effect of inorganic anions

Chloride ions are ubiquitously present in natural water [44]. The presence of chloride ions can accelerate the degradation of TC (Fig. 5a), which can be attributed to the enhanced oxidative ability of the reactive medium through the generation of $Cl^\cdot$ (Eq. (2)) and $Cl_2^{\cdot-}$ radicals (Eq. (3)) [45,46].



The effect of bicarbonate ions on TC degradation was investigated, and the results are shown in Fig. 5b. The degradation efficiency of TC increased from 73.5% to 81.1, 86.0 and 86.8% with the addition of 1.0 mM, 5.0 mM and 10.0 mM $NaHCO_3$, respectively (Fig. 5b). To further study the role of bicarbonate ions, pH variation and the total leached copper ions were also monitored (Fig. 5c and d). The solution pH decreased from 5.81 to 3.22 in the absence of $NaHCO_3$, and decreased from 7.70 to 3.81 in the presence of 1 mM $NaHCO_3$. However, with the addition of 5.0 mM $NaHCO_3$, the solution pH did not change, and the

solution pH gradually increased when 10.0 mM $NaHCO_3$ was present. This phenomenon could be attributed to the buffering ability of $NaHCO_3$. Besides, it should be noted that the leaching of copper ions immensely declined from 102.2 mg/L to 10.8 and 6.9 mg/L in the presence of 5.0 and 10.0 mM $NaHCO_3$, respectively (Fig. 5d). Very recently, Zhou et al. reported that the *in-situ* generated cuprous ions (Cu^+) activated PS to produce $SO^{\cdot-}$ 4, which can degrade the organic pollutants [27]. In this work, although the leaching of copper ions was inhibited when bicarbonate ions were present, the catalytic performance was improved (Fig. 5). This result demonstrated that $Cu^\cdot$ was the active component in $Cu/CuFe_2O_4/PS$ system. $Cu^\cdot$, just like $Fe^\cdot$, can act as the electron bridge to accelerate the electron transfer between PS and the catalyst to facilitate the generation of reactive radicals (Eqs. (4) and (5)) [10,47].



In addition, the effects of some other inorganic anions (CO_3^{2-} , PO_3^{3-} and NO_3^-) on TC degradation in $Cu/CuFe_2O_4/PS$ system were also investigated. The presence of NO_3^- could increase TC removal efficiency (Fig. S7). The $SO^{\cdot-}$ 4radicals can react with NO_3^- to produce $NO_2^\cdot$ radicals, which possess a significant oxidation power [45], resulting in the positive effect on TC degradation. The effect of PO_3^{3-} on TC degradation is almost negligible (Fig. S7). CO_3^{2-} ions can react with $SO^{\cdot-}$ 4or $OH^\cdot$ to form $CO_3^{\cdot-}$ radicals [48]. The oxidation power of $CO_3^{\cdot-}$ radicals is relatively lower compared with $SO^{\cdot-}$ 4or $OH^\cdot$ radicals [49], leading to a slight inhibition on TC removal (Fig. S7).

3.3. Stability and recyclability

The stability and recyclability of $Cu/CuFe_2O_4$ composite were evaluated by its reuse performance (Fig. 6). After every 120 min

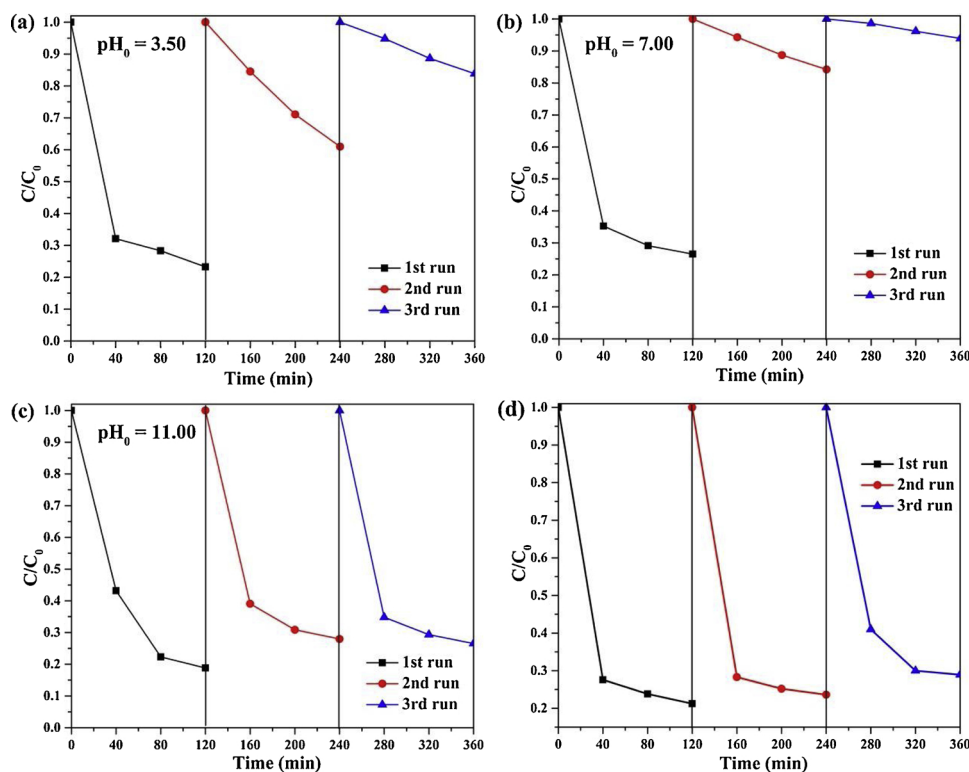


Fig. 6. TC degradation in Cu/CuFe₂O₄/PS system in three successive runs at pH₀ 3.50 (a), 7.00 (b) and 11.00 (c); (d) the cycle tests in presence of NaHCO₃ (10.0 mM). Conditions: [PS]₀ = 1.5 g L⁻¹, [TC] = 50 mg L⁻¹, [Cu/CuFe₂O₄] = 0.3 g L⁻¹.

reaction, the used catalyst was magnetically collected and washed with distilled water, and then dried at 120 °C. After three runs, the degradation efficiency of TC greatly decreased from 76.7% to 16.2% at pH₀ 3.50, and from 73.5% to 6.1% at pH₀ 7.00 (Fig. 6a and b). At pH₀ 3.50, after the first run, almost all of Cu⁺ in Cu/CuFe₂O₄ catalyst was oxidized to copper ions, and only a few copper ions were detected in the second and third runs (Fig. S8a). The similar result was found at pH₀ 7.00 (Fig. S8b). In addition, XPS analysis confirmed that the typical peaks of Cu⁺ in the used catalyst disappeared at pH₀ 3.50 or 7.00 (as shown in Fig. 8). These results indicated that the rapid decrease in activity at pH₀ 3.50 or 7.00 was mainly attributed to the loss of active Cu⁺. When the solution pH₀ was 11.00, the removal efficiency of TC slightly decreased from 81.2% to 73.5% after three runs (Fig. 6c). Although the leaching of copper ions greatly decreased after three runs at pH₀ 11.00 (Fig. S8c), over 70% of TC can still be degraded in the third run. The better recyclability of Cu/CuFe₂O₄ catalyst under alkaline condition can be explained from two aspects: (i) basic condition enhanced the reactivity of PS to form SO^{•-} - 4 radicals [37]; (ii) the remained CuFe₂O₄ can activate PS to degrade TC at alkaline condition (Fig. 3c).

Bicarbonate ions are widely present in natural water and wastewater [34], and the above results confirmed that bicarbonate ions can improve the catalytic activity and inhibit the copper ions leaching. In the presence of NaHCO₃ (10 mM), TC degradation efficiency slightly decreased in Cu/CuFe₂O₄/PS system after three recycling tests (Fig. 6d). Besides, the leaching of copper ions during the oxidation reaction was also monitored in the successive runs (Fig. S8d). To be specific, the leaching of copper ions in the first, second and third runs was 6.9, 3.2 and 2.3 mg/L, respectively. With or without the addition of NaHCO₃, the XPS spectra of Cu 2p for Cu/CuFe₂O₄ catalyst after 120 min reaction are displayed in Fig. S9. Clearly, about 30.5% of Cu⁺ remained in the used Cu/CuFe₂O₄ catalyst (32.6% of Cu⁺ in fresh sample) in the presence of 10 mM NaHCO₃, while Cu⁺ cannot be detected in the used catalyst without the addition of NaHCO₃. These results suggested that the bicarbonate ions greatly suppressed the

leaching of copper ions to keep the stability of Cu/CuFe₂O₄ catalyst, thereby enhancing its recyclability.

3.4. Mechanism study

Previous works have proved that SO^{•-} - 4 radicals can be formed during the catalytic activation of PS by transition metals and their oxides [34,50]. Besides, OH[•] radicals can also be generated through Eqs. (6) and (7) [39]. Herein, the free radical quenching tests were performed to detect SO^{•-} - 4 and OH[•] radicals in Cu/CuFe₂O₄/PS system at different pH₀. Methanol (MA) was used as a mighty quencher of both OH[•] (rate constant 1.0 × 10⁹ M⁻¹ s⁻¹) and SO^{•-} - 4 (2.5 × 10⁷ M⁻¹ s⁻¹) [25], and *t*-butanol (TBA) was a selective scavenger of OH[•] due to the 1000-fold higher rate constant with OH[•] ((3.8–7.6) × 10⁹ M⁻¹ s⁻¹) than with SO^{•-} - 4 ((4.0–9.1) × 10⁵ M⁻¹ s⁻¹) [21].



At pH₀ 3.50, about 72.4% of TC was removed after 80 min without any scavenger addition, and TC degradation efficiency dropped to 57.8% in the presence of TBA due to the quenching of OH[•] radicals (Fig. 7a). The removal efficiency of TC greatly decreased to 23.8% with the addition of MA owing to the trapping of both SO^{•-} - 4 and OH[•] (Fig. 7a). Obviously, the similar trend was observed under neutral or basic condition (Fig. 7b and c). All these findings indicated that both SO^{•-} - 4 and OH[•] radicals played a significant role for TC degradation under different pH conditions. To further confirm the existence of SO^{•-} - 4 and OH[•] radicals, EPR spectroscopy was performed using DMPO as the spin trapping reagent, and the result is displayed in Fig. 7d. The characteristic signals for DMPO - OH (a_N = a_H = 14.9 G) and DMPO - SO₄ adducts (a_N = 13.2 G, a_H = 9.6 G, a_H = 1.48 G) were consistent with the data for SO^{•-} - 4 and OH[•] radicals in the previous works [10,39,51].

The XPS spectra of the used Cu/CuFe₂O₄ were measured to explore

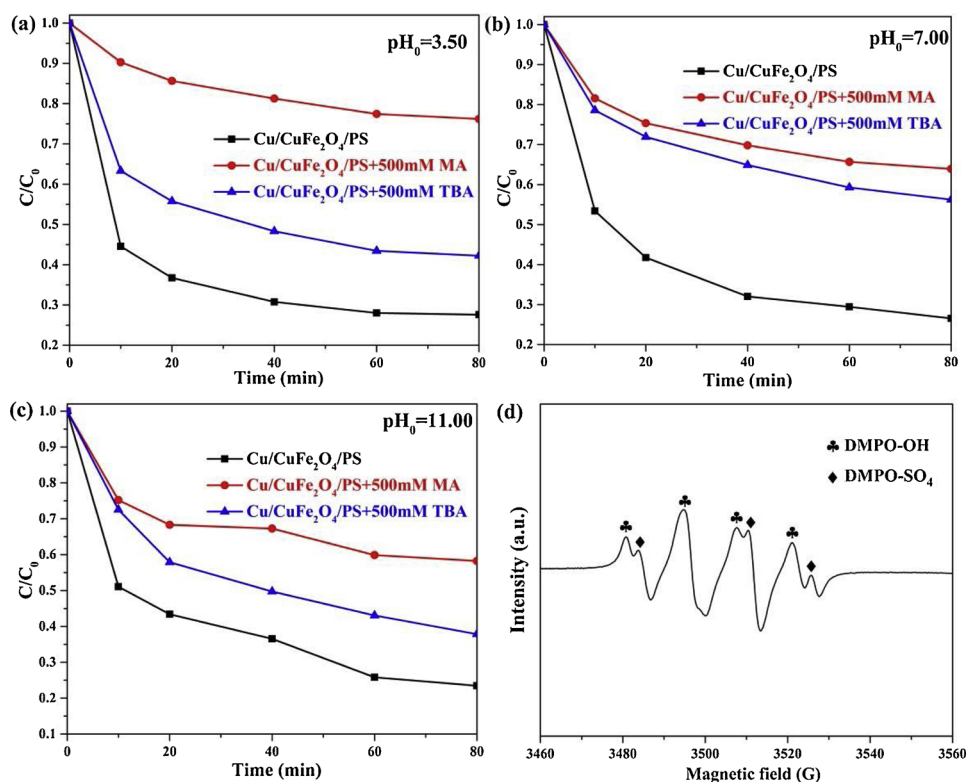


Fig. 7. The effect of different scavengers on TC degradation under different pH_0 : (a) $pH_0 = 3.50$, (b) $pH_0 = 7.00$ and (c) $pH_0 = 11.00$; (d) EPR spectrum in Cu/CuFe₂O₄/PS system ([DMPO]: 10 mM, $pH_0 = 7.00$). Conditions: $[PS]_0 = 1.5 \text{ g L}^{-1}$, $[TC] = 50 \text{ mg L}^{-1}$, $[Cu/CuFe_2O_4] = 0.3 \text{ g L}^{-1}$.

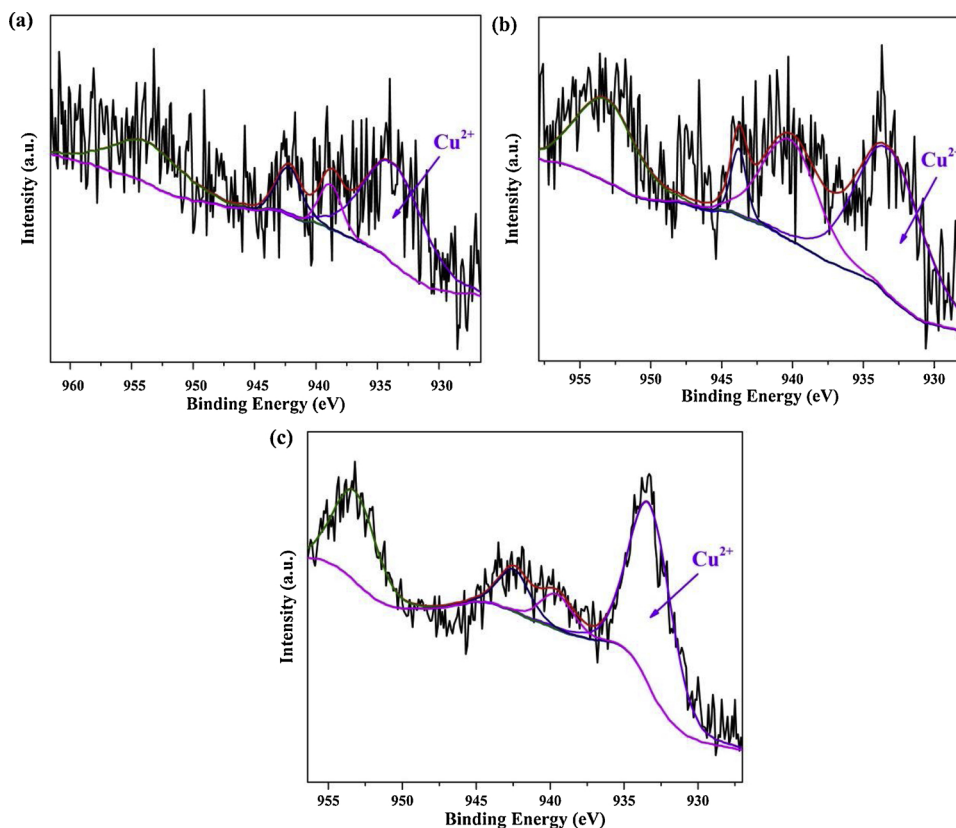


Fig. 8. Cu 2p XPS spectra of the used Cu/CuFe₂O₄ samples at pH_0 3.50 (a); pH_0 7.00 (b) and pH_0 11.00 (c).

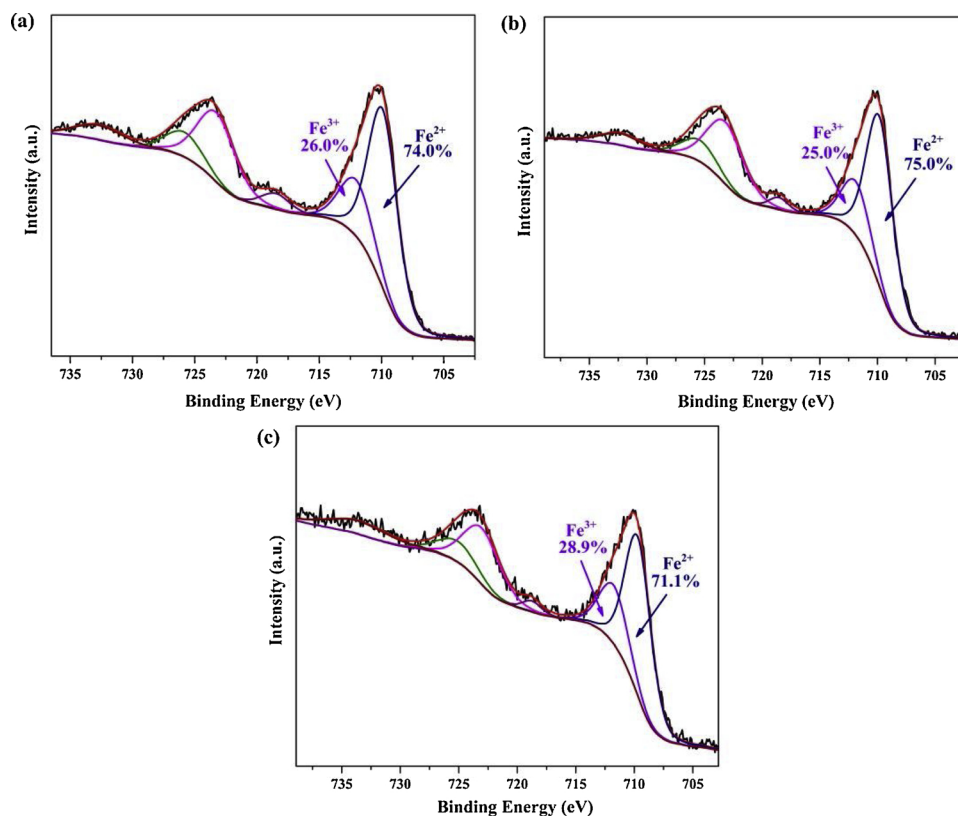
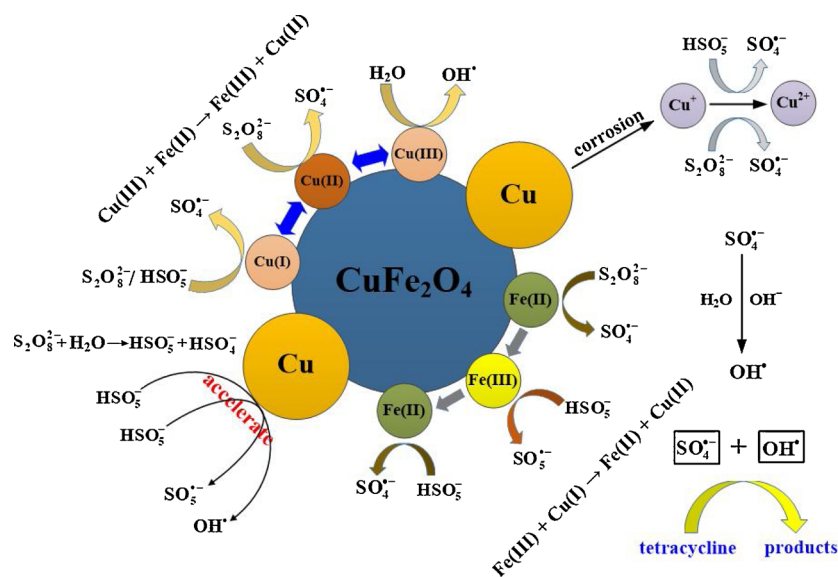


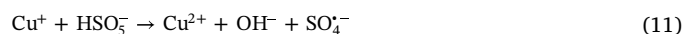
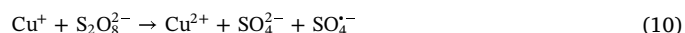
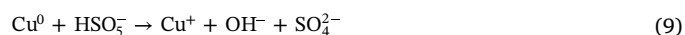
Fig. 9. Fe 2p XPS spectra of the used Cu/CuFe₂O₄ samples at pH₀ 3.50 (a); pH₀ 7.00 (b) and pH₀ 11.00 (c).



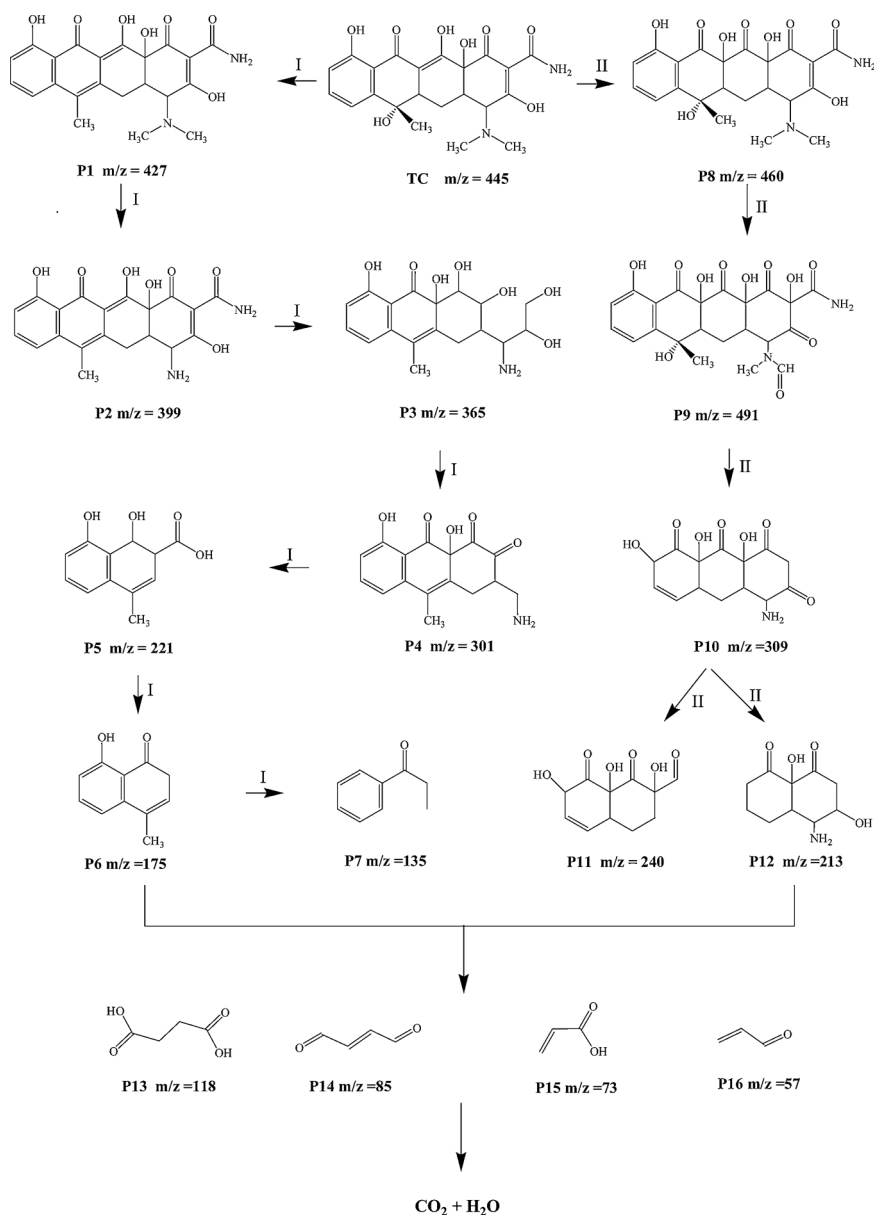
Scheme 1. The reaction mechanism for TC degradation in Cu/CuFe₂O₄/PS system.

the PS activation mechanism (Figs. 8 and 9). The XPS spectra of O 1s for the used catalysts were presented in Fig. S10. The peaks at 951.78 and 931.98 eV belong to Cu⁰ were disappeared in the used catalysts (Fig. 8), suggesting that the corrosion of Cu⁰ actually occurred under different pH₀ conditions. Although Cu⁰ in this composite catalyst was totally dissolved after 120 min of reaction, the role of Cu⁰ on the catalytic performance in the early stage cannot be ignored according to the above analysis. The in-situ generated Cu⁺ ions (Eqs. (8) and (9)) could activate PS to produce SO₄^{•-} radicals (Eqs. (10) and (11)) [25,27]. To prove this point, the effect of neocuproine hemihydrate (NCP) on TC degradation was investigated to reveal the role of Cu⁺ in Cu/CuFe₂O₄/

PS system because NCP is an effective chelating agent for Cu⁺ [27]. Clearly, TC removal efficiencies decreased when NCP was added under different pH conditions (Fig. S11).



As displayed in Figs. 2 and 8, the negative migration of the peaks



assigned to Cu(II) was observed from 933.71 eV to 933.20, 933.18 and 933.68 eV, respectively, indicating that Cu(I) was generated via Eq. (12) under different pH conditions [14]. The in-situ generated Cu(I) on the surface of CuFe₂O₄ could react with PS to produce SO[•]−4 radicals through Eqs. (13) and (14) [14]. In addition, Cu(II) on the surface of CuFe₂O₄ could also be oxidized to Cu(III) by the adsorbed PS (Eq. (15)), and the in-situ generated Cu(III) could react with H₂O to produce OH[•] radicals (Eq. (16)) [14,21].

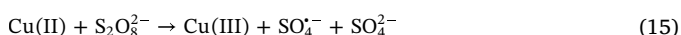
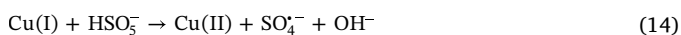
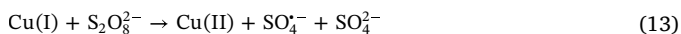
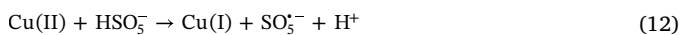
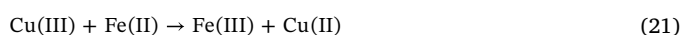
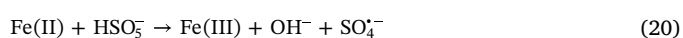
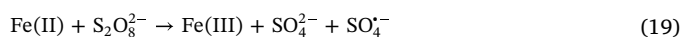
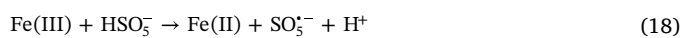


Fig. 9 presents the XPS spectra of Fe 2p regions for the used Cu/CuFe₂O₄. Compared with fresh catalyst, the percentage of Fe(II) in the used Cu/CuFe₂O₄ increased from 63.3% to 74.0 (pH₀ 3.50), 75.0 (pH₀

7.00) and 71.1% (pH₀ 11.00), respectively. This result was due to the reduction of Fe(III) by Cu(I) via Eq. (17) [52] or the reaction between Fe(III) and HSO[•]−5 through Eq. (18) [14]. The Fe(II) species could react with PS to generate SO[•]−4 radicals via Eqs. (19) and (20) [9,21]. Besides, the oxidation of Fe(II) to Fe(III) by Cu(III) can also achieved (Eq. (21)) [14].



Based on the results obtained above, a possible mechanism for TC degradation in Cu/CuFe₂O₄/PS system was proposed (Scheme 1). PS can react with H₂O to produce HSO[•]−5 and HSO[•]−4 (Eq. (1)), and Cu[•] could act as the electron bridge to accelerate electron transfer between PS molecules and Cu/CuFe₂O₄ catalyst (Eqs. (4) and (5)). Besides, Cu[•]

could be oxidized by PS to release Cu^+ ions (Eqs. (8) and (9)), which can activate PS to produce $\text{SO}^\bullet - 4$ radicals (Eqs. (10) and (11)). Cu(I) could be generated through Eq. (12). Cu(I) and Cu(II) on the surface of CuFe_2O_4 could react with PS or $\text{HSO}^- 5$ to form $\text{SO}^\bullet - 4$ radicals (Eqs. (13)–(15)), and the formed Cu(III) can react with H_2O to generate $\text{OH}^\bullet$ radicals (Eq. (16)). Fe(III) on the surface of CuFe_2O_4 can react with Cu(I) (Eq. (17)) or $\text{HSO}^- 5$ (Eq. (18)) to produce Fe(II) species, which can also activate PS or $\text{HSO}^- 5$ to generate $\text{SO}^\bullet - 4$ radicals (Eqs. (19) and (20)). Moreover, the recovery of Fe(III) and Cu(II) could also be achieved through the reaction between Cu(III) and Fe(II) (Eq. (21)). Furthermore, $\text{SO}^\bullet - 4$ radicals can react with H_2O or OH^- to produce $\text{OH}^\bullet$ radicals (Eqs. (6) and (7)). Finally, the in-situ generated $\text{SO}^\bullet - 4$ and $\text{OH}^\bullet$ radicals could oxidize TC molecules.

3.5. Possible degradation pathways of TC

HPLC-MS/MS was employed to detect the degradation intermediates of TC in Cu/CuFe₂O₄/PS system, and the results are shown in Fig. S12. The possible molecular structures of the intermediates are listed in Table S1, and two possible degradation pathways of TC were proposed in Scheme 2. For the pathway I, P1 with m/z 427 was generated from the dehydration process of TC [7], which was also proved by Liu et al. [53]. The intermediate with m/z of 399 (P2) was formed through the N-demethylation reaction of P1 [54], and the ring-opening products including P3 (m/z 365), P4 (m/z 301), P5 (m/z 221), P6 (m/z 175) and P7 (m/z 135) were assigned as the further oxidation products of P2 by $\text{OH}^\bullet$ and $\text{SO}^\bullet - 4$ attack [5,55]. In terms of pathway II, the intermediate P8 with m/z 460 was identified as the hydroxylation product of TC [6], and the intermediate P9 with m/z 491 was generated from P8 by $\text{OH}^\bullet$ radical attack [53]. Then, along with the ring open, P10 (m/z 309), P11 (m/z 240) and P12 (m/z 213) were formed via the decarbonylation, dealkylation, dehydration and dehydroxylation processes [6,55,56], respectively. In addition, the smaller molecule intermediates, including P13 (m/z 118), P14 (m/z 85), P15 (m/z 73) and P16 (m/z 57), could be formed from the further oxidation of the intermediates in pathway I (P7) and pathway II (P11 and P12) by $\text{OH}^\bullet$ and $\text{SO}^\bullet - 4$ attack [5,57]. Finally, some intermediates in Cu/CuFe₂O₄/PS system can be further oxidized into CO_2 and H_2O . This result can be supported by the TOC removal shown in Fig. 4b.

4. Conclusions

In summary, the as-prepared magnetic Cu/CuFe₂O₄ catalyst showed an excellent performance to active PS for TC degradation over a wide pH range. In Cu/CuFe₂O₄/PS system, the removal efficiency of TC is slightly increased as PS concentration or catalyst dosage increases but drops at a certain PS concentration (2.25 g/L) or catalyst dosage (0.4 g/L). Elevated temperature was found to promote the decomposition of TC. The addition of inorganic anions also has impacts on TC removal in Cu/CuFe₂O₄/PS system. For example, the ions such as Cl^- , $\text{HCO}^- 3$ and $\text{NO}^- 3$ were found to facilitate the degradation efficiency of TC, while the ions like $\text{CO}_2 - 3$ and $\text{PO}_3 - 4$ have slight suppression effect on TC degradation. The stability and recyclability of Cu/CuFe₂O₄ composite could be enhanced in the presence of bicarbonate ions. Both $\text{SO}^\bullet - 4$ and $\text{OH}^\bullet$ radicals were considered as the dominant active species in Cu/CuFe₂O₄/PS system. Two possible degradation pathways of TC were proposed based on the detected intermediates.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhazmat.2019.03.075>.

References

- [1] W.L. Shi, F. Guo, S.L. Yuan, In situ synthesis of Z-scheme $\text{Ag}_3\text{PO}_4/\text{CuBi}_2\text{O}_4$ photocatalysts and enhanced photocatalytic performance for the degradation of tetracycline under visible light irradiation, *Appl. Catal. B: Environ.* 209 (2017) 720–728.
- [2] S. Leong, D. Li, K. Hapgood, X.W. Zhang, H.T. Wang, $\text{Ni}(\text{OH})_2$ decorated rutile TiO_2 for efficient removal of tetracycline from wastewater, *Appl. Catal. B: Environ.* 198 (2016) 224–233.
- [3] I. Godos, R. Munoz, B. Guieysse, Tetracycline removal during wastewater treatment in high-rate algal ponds, *J. Hazard. Mater.* 229–230 (2012) 446–449.
- [4] H. Kim, Y. Hong, J.E. Park, V.K. Sharma, S.I. Cho, Sulfonamides and tetracyclines in livestock wastewater, *Chemosphere* 91 (2013) 888–894.
- [5] Z.J. Xie, Y.P. Feng, F.L. Wang, D.N. Chen, Q.X. Zhang, Y.Q. Zeng, W.Y. Lv, G.G. Liu, Construction of carbon dots modified $\text{MoO}_3/\text{g-C}_3\text{N}_4$ Z-scheme photocatalyst with enhanced visible-light photocatalytic activity for the degradation of tetracycline, *Appl. Catal. B: Environ.* 229 (2018) 96–104.
- [6] J.Y. Cao, Z.K. Xiong, B. Lai, Effect of initial pH on the tetracycline (TC) removal by zero-valent iron: adsorption, oxidation and reduction, *Chem. Eng. J.* 343 (2018) 492–499.
- [7] Y.Y. Chen, Y.L. Ma, J. Yang, L.Q. Wang, J.M. Lv, C.J. Ren, Aqueous tetracycline degradation by H_2O_2 alone: removal and transformation pathway, *Chem. Eng. J.* 307 (2017) 15–23.
- [8] Y.B. Ding, L.H. Zhu, N. Wang, H.Q. Tang, Sulfate radicals induced degradation of tetrabromobisphenol A with nanoscaled magnetic CuFe_2O_4 as a heterogeneous peroxymonosulfate, *Appl. Catal. B: Environ.* 129 (2013) 153–162.
- [9] Q.Q. Ji, J. Li, Z.K. Xiong, B. Lai, Enhanced reactivity of microscale Fe/Cu bimetallic particles (mFe/Cu) with persulfate (PS) for p-nitrophenol (PNP) removal in aqueous solution, *Chemosphere* 172 (2017) 10–20.
- [10] X.G. Duan, H.Q. Sun, J. Kang, Y.X. Wang, S. Indrawirawan, S.B. Wang, Insights into heterogeneous catalysis of persulfate activation on dimensional-structured nanocarbons, *ACS Catal.* 5 (2015) 4629–4636.
- [11] H. Zhong, Y.L. Tian, Q. Yang, M.L. Brusseau, L. Yang, G.M. Zeng, Degradation of landfill leachate compounds by persulfate for groundwater remediation, *Chem. Eng. J.* 307 (2017) 399–407.
- [12] X. Jiang, Y.H. Guo, L.B. Zhang, W.J. Jiang, R.Z. Xie, Catalytic degradation of tetracycline hydrochloride by persulfate activated with nano Fe^0 immobilized mesoporous carbon, *Chem. Eng. J.* 341 (2018) 392–401.
- [13] A.J. Jafari, B. Kakavandi, N. Jaafarzadeh, R.R. Kalantary, M. Ahmadi, A.A. Babaei, Fenton-like catalytic oxidation of tetracycline by $\text{AC}/\text{Fe}_3\text{O}_4$ as a heterogeneous persulfate activator: adsorption and degradation studies, *J. Ind. Eng. Chem.* 45 (2017) 323–333.
- [14] J. Li, Y. Ren, F.Z. Ji, B. Lai, Heterogeneous catalytic oxidation for the degradation of p-nitrophenol in aqueous solution by persulfate activated with CuFe_2O_4 magnetic nano-particles, *Chem. Eng. J.* 324 (2017) 63–73.
- [15] N. Nasrallah, M. Kebir, Z. Koudri, M. Trari, Photocatalytic reduction of Cr(VI) on the novel hetero-system $\text{CuFe}_2\text{O}_4/\text{CdS}$, *J. Hazard. Mater.* 185 (2011) 1398–1404.
- [16] P.P. Jing, J.N. Li, L.N. Pan, J.B. Wang, X.J. Sun, Q.F. Liu, Efficient photocatalytic degradation of acid fuchsin in aqueous solution using separate porous tetragonal- CuFe_2O_4 nanotubes, *J. Hazard. Mater.* 284 (2015) 163–170.
- [17] T. Zhang, H.B. Zhu, J.P. Croue, Production of sulfate radical from peroxymonosulfate induced by a magnetically separable CuFe_2O_4 spinel in water: efficiency, stability, and mechanism, *Environ. Sci. Technol.* 47 (2013) 2784–2791.
- [18] Y. Xu, J. Ai, H. Zhang, The mechanism of degradation of bisphenol A using the magnetically separable CuFe_2O_4 /peroxymonosulfate heterogeneous oxidation process, *J. Hazard. Mater.* 309 (2016) 87–96.
- [19] Y.H. Guan, J. Ma, Y.M. Ren, Y.L. Liu, J.Y. Xiao, L.Q. Lin, C. Zhang, Efficient degradation of atrazine by magnetic porous copper ferrite catalyzed peroxymonosulfate oxidation via the formation of hydroxyl and sulfate radicals, *Water Res.* 47 (2013) 5431–5438.
- [20] Y.T. Zhao, G.Y. He, W. Dai, H.Q. Chen, High catalytic activity in the phenol hydroxylation of magnetically separable CuFe_2O_4 -Reduced graphene oxide, *Ind. Eng. Chem. Res.* 53 (2014) 12566–12574.
- [21] X.L. Zhang, M.B. Feng, R.J. Qu, H. Liu, L.S. Wang, Z.Y. Wang, Catalytic degradation of diethyl phthalate in aqueous solution by persulfate activated with nano-scaled magnetic CuFe_2O_4 /MWCNTs, *Chem. Eng. J.* 301 (2016) 1–11.
- [22] I. Hussain, M.Y. Li, Y.Q. Zhang, S.B. Huang, W. Hayat, Y.C. Li, X.D. Du, G.Q. Liu, Efficient oxidation of arsenic in aqueous solution using zero valent iron-activated persulfate process, *J. Environ. Chem. Eng.* 5 (2017) 3983–3990.
- [23] J. Khatri, P.V. Nidheesh, T.S. Anantha Singh, M. Suresh Kumar, Advanced oxidation processes based on zero-valent aluminium for treating textile wastewater, *Chem. Eng. J.* 348 (2018) 67–73.
- [24] B.B. Huang, W.T. Qian, C.X. Yu, T. Wang, G.M. Zeng, C. Lei, Effective catalytic hydrodechlorination of o-, p- and m-chloronitrobenzene over Ni/Fe nanoparticles: effects of experimental parameter and molecule structure on the reduction kinetics and mechanisms, *Chem. Eng. J.* 306 (2016) 607–618.
- [25] P. Zhou, J. Zhang, Y.L. Zhang, G.C. Zhang, W.S. Li, C.M. Wei, J. Liang, Y. Liu, S.H. Shu, Degradation of 2,4-dichlorophenol by activating persulfate and peroxymonosulfate using micron or nanoscale zero-valent copper, *J. Hazard. Mater.* 344 (2018) 1209–1219.

- [26] G. Wen, S.J. Wang, J. Ma, T.L. Huang, Z.Q. Liu, L. Zhao, J.L. Xu, Oxidative degradation of organic pollutants in aqueous solution using zero valent copper under aerobic atmosphere condition, *J. Hazard. Mater.* 275 (2014) 193–199.
- [27] P. Zhou, J. Zhang, J.L. Liu, Y.L. Zhang, J. Liang, Y. Liu, B. Liu, W. Zhang, Degradation of organic contaminants by activated persulfate using zero valent copper in acidic aqueous conditions, *RSC Adv.* 6 (2016) 99532–99539.
- [28] Z.K. Xiong, B. Lai, P. Yang, Y.X. Zhou, J.L. Wang, S.P. Fang, Comparative study on the reactivity of Fe/Cu bimetallic particles and zero valent iron (ZVI) under different conditions of N₂, air or without aeration, *J. Hazard. Mater.* 297 (2015) 261–268.
- [29] B. Lai, Y.H. Zhang, Z.Y. Chen, P. Yang, Y.X. Zhou, J.L. Wang, Removal of p-nitrophenol (PNP) in aqueous solution by the micron-scale iron–copper (Fe/Cu) bi-metallic particles, *Appl. Catal. B: Environ.* 144 (2014) 816–830.
- [30] Z.L. Li, J.C. Lyu, M. Ge, Synthesis of magnetic Cu/CuFe₂O₄ nanocomposite as a highly efficient Fenton-like catalyst for methylene blue degradation, *J. Mater. Sci.* 53 (2018) 15081–15095.
- [31] Y.C. Dong, Y.S. Chui, R.G. Ma, C.W. Cao, H. Cheng, Y.Y. Li, J.A. Zapien, One-pot scalable synthesis of Cu-CuFe₂O₄/graphene composites as anode materials for lithium-ion batteries with enhanced lithium storage properties, *J. Mater. Chem. A* 2 (2014) 13892–13897.
- [32] X.Y. Zhang, Y.B. Ding, H.Q. Tang, X.Y. Han, L.H. Zhu, N. Wang, Degradation of bisphenol A by hydrogen peroxide activated with CuFeO₂microparticles as a heterogeneous Fenton-like catalyst: Efficiency, stability and mechanism, *Chem. Eng. J.* 236 (2014) 251–262.
- [33] Y.T. Zhang, C. Liu, B.B. Xu, F. Qi, W. Chu, Degradation of benzotriazole by a novel Fenton-like reaction with mesoporous Cu/MnO₂: combination of adsorption and catalysis oxidation, *Appl. Catal. B: Environ.* 199 (2016) 447–457.
- [34] Y. Lei, C.S. Chen, Y.J. Tu, Y.H. Huang, H. Zhang, Heterogeneous degradation of organic pollutants by persulfate activated by CuO-Fe₃O₄: mechanism, stability, and effects of pH and bicarbonate ions, *Environ. Sci. Technol.* 49 (2015) 6838–6845.
- [35] T.T. Li, S. Cao, C. Yang, Y. Chen, X.J. Lv, W.F. Fu, Electrochemical water oxidation by in situ-generated copper oxide film from [Cu(TEOA)(H₂O)₂][SO₄] complex, *Inorg. Chem.* 54 (2015) 3061–3067.
- [36] S.T. Xing, Z.C. Zhou, Z.C. Ma, Y.S. Wu, Characterization and reactivity of Fe₃O₄/FeMnO_x core/shell nanoparticles for methylene blue discoloration with H₂O₂, *Appl. Catal. B: Environ.* 107 (2011) 386–392.
- [37] O.S. Furman, A.L. Teel, R.J. Watts, Mechanism of base activation of persulfate, *Environ. Sci. Technol.* 44 (2010) 6423–6428.
- [38] D.H. Ding, C. Liu, Y.F. Ji, Q. Yang, L.L. Chen, C.L. Jiang, T.M. Cai, Mechanism insight of degradation of norfloxacin by magnetite nanoparticles activated persulfate: identification of radicals and degradation pathway, *Chem. Eng. J.* 308 (2017) 330–339.
- [39] A. Jawad, J. Lang, Z.W. Liao, A. Khan, J. Iftikhar, Z.A. Lv, S.J. Long, Z.L. Chen, Z.Q. Chen, Activation of persulfate by CuO_x@Co-LDH: A novel heterogeneous system for contaminant degradation with broad pH window and controlled leaching, *Chem. Eng. J.* 335 (2018) 548–559.
- [40] Z. Wang, D.Y. Deng, L.L. Yang, Degradation of dimethyl phthalate in solutions and soil slurries by persulfate at ambient temperature, *J. Hazard. Mater.* 271 (2014) 202–209.
- [41] R.E. Huie, C.L. Clifton, Rate constants for hydrogen abstraction reactions of the sulfate radical, SO₄^{•−}. Alkanes and ethers, *Int. J. Chem. Kinet.* 21 (1989) 611–619.
- [42] M.B. Peng, R.J. Qu, X.L. Zhang, P. Sun, Y.X. Sui, L.S. Wang, Z.Y. Wang, Degradation of flumequine in aqueous solution by persulfate activated with common methods and polyhydroquinone-coated magnetite/multi-walled carbon nanotubes catalysts, *Water Res.* 85 (2015) 1–10.
- [43] D.Y. Deng, L.B. Peng, M.Y. Guan, Y. Kang, Impact of activation methods on persulfate oxidation of methyl tert-butyl ether, *J. Hazard. Mater.* 264 (2014) 521–528.
- [44] S. Wacławek, H.V. Lutze, K. Grübel, V.V.T. Padil, M. Černík, D.D. Dionysiou, Chemistry of persulfates in water and wastewater treatment: a review, *Chem. Eng. J.* 330 (2017) 44–62.
- [45] A. Ghauch, A. Baalbaki, M. Amasha, R.E. Asmar, O. Tantawi, Contribution of persulfate in UV-254 nm activated systems for complete degradation of chloramphenicol antibiotic in water, *Chem. Eng. J.* 317 (2017) 1012–1025.
- [46] C.Q. Tan, D.F. Fu, N.Y. Gao, Q.D. Qin, Y. Xu, H.M. Xiang, Kinetic degradation of chloramphenicol in water by UV/persulfate system, *J. Photochem. Photobiol. A* 332 (2017) 406–412.
- [47] H.M. Sun, X.J. Yang, L.J. Zhao, T.H. Xu, J.S. Lian, One-pot hydrothermal synthesis of octahedral CoFe/CoFe₂O₄ submicron composite as heterogeneous catalysts with enhanced peroxymonosulfate activity, *J. Mater. Chem. A* 4 (2016) 9455–9465.
- [48] J. Deng, Y.S. Shao, N.Y. Gao, Y. Deng, S.Q. Zhou, X.H. Hu, Thermally activated persulfate (TAP) oxidation of antiepileptic drug carbamazepine in water, *Chem. Eng. J.* 228 (2013) 765–771.
- [49] G.V. Buxton, N.D. Wood, S. Dyster, Ionisation Constants of [•]OH and HO[•] 2 in Aqueous Solution up to 200°C, *J. Chem. Soc. Faraday Trans. 1* (84) (1988) 1113–1121.
- [50] H.Y. Liang, Y.Q. Zhang, S.B. Huang, I. Hussain, Oxidative degradation of p-chloroaniline by copper oxidate activated persulfate, *Chem. Eng. J.* 218 (2013) 384–391.
- [51] C.D. Qi, X.T. Liu, J. Ma, C.Y. Lin, X.X. Li, H.J. Zhang, Activation of peroxymonosulfate by base: implications for the degradation of organic pollutants, *Chemosphere* 151 (2016) 280–288.
- [52] Q.D. Qin, Y.H. Liu, X.C. Li, T. Sun, Y. Xu, Enhanced heterogeneous Fenton-like degradation of methylene blue by reduced CuFe₂O₄, *RSC Adv.* 8 (2018) 1071–1077.
- [53] Y.Q. Liu, X.X. He, Y.S. Fu, D.D. Dionysiou, Kinetics and mechanism investigation on the destruction of oxytetracycline by UV-254 nm activation of persulfate, *J. Hazard. Mater.* 305 (2016) 229–239.
- [54] X.N. Wang, J.P. Jia, Y.L. Wang, Combination of photocatalysis with hydrodynamic cavitation for degradation of tetracycline, *Chem. Eng. J.* 315 (2017) 274–282.
- [55] T.Y. Wang, W. Quan, D.L. Jiang, L.L. Chen, D. Li, S.C. Meng, M. Chen, Synthesis of redox-mediator-free direct Z-scheme AgI/WO₃ nanocomposite photocatalysts for the degradation of tetracycline with enhanced photocatalytic activity, *Chem. Eng. J.* 300 (2016) 280–290.
- [56] D.L. Jiang, T.Y. Wang, Q. Xu, D. Li, S.C. Meng, M. Chen, Perovskite oxide ultrathin nanosheets/g-C₃N₄ 2D-2D heterojunction photocatalysts with significantly enhanced photocatalytic activity towards the photodegradation of tetracycline, *Appl. Catal. B: Environ.* 201 (2017) 617–628.
- [57] N. Barhoumi, H. Olvera-Vargas, N. Oturan, D. Huguenot, A. Gadri, S. Ammar, E. Brillas, M.A. Oturan, Kinetics of oxidative degradation/mineralization pathways of the antibiotic tetracycline by the novel heterogeneous electro-Fenton process with solid catalyst chalcocopyrite, *Appl. Catal. B: Environ.* 209 (2017) 637–647.